

Globalisation, Income, and Welfare: Imported Goods Variety and Quality*

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This version: July 2, 2026

Abstract: Did globalisation benefit consumers in poor countries? Using detailed datasets on goods trade for up to 138 countries, we measure the welfare gains from access to greater variety and increased quality of imported goods. The variety gains are large, a median 15 percent decline in the cost of living, and strongly pro-poor, an effect size in poorer countries over double that of richer ones. Because these gains accrue from falling trade costs and not increasing income, they made poor countries richer in real consumption even in the absence of income growth. Quality, in contrast, follows income growth and barely converged.

Keywords: gains from trade, import variety, globalisation, convergence, consumption

JEL codes: F14, F61, O19, D63

*In preparing this paper, the authors used an AI-based assistant tool (Anthropic's Claude) to support coding, the construction of some of the data, the generation of tables and figures, and language editing. The research design, the empirical analysis and its specification, the drafting and writing of the manuscript, the interpretation of the results, and all conclusions are the authors' own. The usual disclaimers apply. Correspondence: Rodolphe Desbordes, SKEMA Business School Paris, 5 Quai Marcel Dassault, 92150 Suresnes, France; email: rodolphe.desbordes@skema.edu

1 Introduction

“Foreign trade. . . [is] highly beneficial to a country, as it increases the amount and variety of the objects on which revenue may be expended.”

David Ricardo, *On the Principles of Political Economy and Taxation* (1817, Ch. 7)

The political economy of globalisation has been written almost entirely on the production side: import competition displaced manufacturing workers (the ‘China shock’; [Autor et al. 2013, 2016](#), [Pierce and Schott 2016](#)), and the gains and losses from trade fell unevenly across regions, sectors, and skills ([Helpman et al. 2016](#), [Dorn and Levell 2024](#), [Rodrik 2024](#)). This is the globalisation that produced a ‘popular’ backlash, and which has been studied very widely not just in its socio-economic but also political implications. A separate literature has quantified the effects of globalisation, in the guise of increased exports and more export destinations, on exporters and on the countries hosting them ([Melitz 2003](#), [Eaton et al. 2011](#), [Brambilla et al. 2012](#), [De Loecker 2013](#), [Arkolakis et al. 2021](#)). But, as David Ricardo noted over 200 years ago, trade also reaches people as *consumers*, through lower prices and access to a wider range and/or quality of goods. On this consumer margin, [Broda and Weinstein \(2006\)](#) estimated that the United States gained on the order of 3 percent of GDP over three decades (1970–2001) from a wider import variety. Their estimate, and most of what we know about consumer gains, applies to rich economies ([Feenstra 2018](#), [Lo Forte 2025](#)).¹

What did the consumer side of globalisation look like for the rest of the world? Which countries gained from increased variety and quality? These questions represent the between-country counterpart of the within-country ‘China shock’ debate. Their answers have the potential to substantially change the progressivity we assigned to the distributional record of trade and globalisation in contemporary and historical periods: if consumer benefits accrued mainly to already-rich economies, global integration widened the gap between nations; if benefits reached poor countries, the gap has narrowed. In this paper we measure the welfare gains from imported variety and quality using HS6 data for 138 countries over 1995–2023 and ask how the respective gains are distributed between richer and poorer economies. Additional analysis uses less granular data (SITC-4) but allows for an extended time horizon (1962–2023).

We find that the variety gains were large and strongly progressive. Applying the [Feenstra \(1994\)](#) exact price index, with substitution elasticities estimated directly product by product from global trade flows, we find a median welfare gain worth about 15 percent of the cost of living. Poorer countries gained roughly twice as much as richer ones on a conservative estimate, and three to four times as much in the baseline results, which include small and possibly suspect trade flows. The gradient survives restricting the analysis to consumer goods or to manufactures (in fact, welfare gains *increase* in both cases), excluding entrepot exporters, alternative weightings, and discarding the small flows that could reflect changes in reporting coverage. Conditioning on initial variety identifies the mechanism: countries

¹Exceptions include [Arkolakis et al. \(2008\)](#), who study Costa Rica, finding only modest gains from variety on welfare, due to small quantities of the new varieties imported.

that began being variety-poor caught up over time, and the originally variety-deprived were disproportionately poor. This is convergence in variety access, the extensive-margin analogue of income convergence.

This gain from variety accrues to real consumption without requiring income growth. It is a fall in the cost of living driven by falling trade costs, not by a country's own income, and conventional income statistics, built on price indices that miss new varieties, do not record it. The point is sharpest once China and India, which witnessed strong income growth, are set aside: between-country incomes barely converged at all over 1995–2023 for the remaining 136 economies, yet the variety channel narrowed real-consumption gaps as much as ever. Poor countries became richer in what they consume without becoming richer in income. In our extension we use SITC-4 data from The Atlas of Economic Complexity and study ten-year windows over 1962 and 2023: we find that convergence in variety access is concentrated in the 1990s and 2000s, dropping to virtually zero in the most recent period.

Did the convergence process also extend to quality? Variety is the extensive margin; quality, whether one buys the dearer or the cheaper version of each good, is the intensive one. We measure it with ϕ , a trade-revealed index of the consumption quality tier built from the within-product unit values of everyday-consumption imports (set out in detail in Section 3). The contrast is sharp: across a range of alternative ways to build the consumption quality index, we find that quality did *not* converge, because it follows income and income barely converged during our time period (excepting India and China). Changes in the quality levels of imports did occur, but these were not conducive to bringing welfare gains to poorer countries. On average, poor countries experienced more choice, but not more quality.

The paper makes three contributions. First, it takes the gains-from-variety accounting of Feenstra (1994) and Broda and Weinstein (2006), applied mostly to individual economies, to the full cross-country distribution. The closest precedent, Fajgelbaum and Khandelwal (2016), measures unequal gains *across consumers within* a country; we measure them *across* countries. Second, it pairs the variety margin with a trade-revealed quality tier and traces their contrasting fates to a simple non-homothetic framework: access follows trade costs, quality follows income. Third, it reads variety access as an extensive-margin counterpart to income β -convergence (Patel et al. 2021, Kremer et al. 2022, Johnson and Papageorgiou 2020): on this single channel, integration narrowed between-country inequality, a progressive consumer-side force the production-side account of the globalisation backlash leaves out. The paper further builds on the measurement of heterogeneous price indices (Hottman et al. 2016), the consumer gains from integration in developing countries (Atkin et al. 2018), the extensive margin of trade (Hummels and Klenow 2005), and the welfare-gains-from-trade framework (Arkolakis et al. 2012). Methodologically, our contribution is to place two consumer margins on a common footing: we combine the exact Feenstra variety price index, with substitution elasticities estimated good by good, with a within-product unit-value index of the quality tier, and apply both to the same bilateral trade records for 138 countries. This lets us ask not only how large the consumer gains from integration were but how they were *distributed*

across rich and poor countries, and whether the breadth and the quality of consumption converged together or apart. We do not claim to conduct the first cross-country variety measurement; however, the distributional incidence and the variety-versus-quality contrast are, to our knowledge, new.

The remainder of the paper lays out a framework for the two margins (Section 2) and sets out the data and the two indices (Section 3). Section 4 establishes the variety gains in three steps: their pro-poor incidence, their aggregate magnitude and effect on between-country inequality, and their trajectory over a sixty-year horizon using an alternative data source. Section 5 contrasts them with the flat quality margin. Section 6 concludes.

2 Two margins, two drivers

A simple framework organises the two margins and shows why they should respond to different forces. Consider a consumer in importer j with real expenditure y_j who buys differentiated varieties of each good. A variety from exporter e has quality z_e and reaches j at the delivered price $\tau_{je} p_e$, where $\tau_{je} \geq 1$ is an iceberg trade cost. Globalisation is a fall in τ .

The extensive margin: access. A variety is consumed only if it is worth importing. With a fixed cost of sourcing each variety, e enters j 's basket only when its delivered price is low enough to cover that cost, that is, when it clears a reservation level,

$$\tau_{je} p_e \leq \bar{p}_{jg}, \quad (1)$$

so the set of accessible varieties $N_{jg}(\tau)$ widens as τ falls and is bounded above by the world set of producers. Two things follow. Lower trade costs raise access, $\partial N / \partial \tau < 0$; and because access is bounded, economies that began with high trade costs, and so few varieties, gain proportionally the most as τ falls, so N_{jg} converges across countries as trade costs converge. The force is τ , not income: at a common trade cost a poor and a rich country reach almost the same set, with income entering only through second-order market-size effects. The Feenstra variety term of Section 3 measures the welfare value of this expanding variety of goods.

The intensive margin: quality. Within the accessible set the consumer chooses a quality tier. Let preferences be non-homothetic in quality, so that the marginal valuation of quality rises with income and the chosen tier q_{jg}^* satisfies

$$\frac{\partial q_{jg}^*}{\partial y_j} > 0, \quad \frac{\partial q_{jg}^*}{\partial \tau_{je}} \approx 0. \quad (2)$$

Quality is a superior characteristic, so the tier rises with income; and because an iceberg cost scales the price of every quality equally, it leaves the quality choice essentially unchanged. The index ϕ of Section 3 is the empirical counterpart of quality tier q^* .

The prediction. A fall in τ raises access for everyone and equalises it, while leaving the quality tier where income puts it. Convergence in access without convergence in quality is therefore the signature of a trade-cost decline under non-homothetic preferences: the extensive margin follows trade costs, the intensive margin follows income. The rest of the paper measures the two margins and finds exactly this. Non-homothetic demand for quality is the mechanism in [Hallak \(2006\)](#), [Fieler \(2011\)](#) and [Fajgelbaum and Khandelwal \(2016\)](#); the extensive-margin response to trade costs is [Hummels and Klenow \(2005\)](#) and [Feenstra \(1994\)](#). The contribution here is to measure both margins jointly across the development spectrum and show that their drivers diverge.

3 Data and measurement

Trade data. The data for our main results are taken from BACI ([Gaulier and Zignago 2010](#)), CEPII's reconciliation of UN Comtrade bilateral flows, in HS92 over the 1995 to 2023 period (around 5,000 product categories). For the purposes of our study, a *variety*, in the sense of [Broda and Weinstein \(2006\)](#), is a country-product pair: a given HS6 good supplied by a given exporter. For each importer we observe the value and quantity of each variety it buys in each year. In our baseline results, we take three-year windows at each end, 1995 to 1997 and 2021 to 2023, to damp year-to-year noise in variety entry and exit. Robustness checks use the data for sub-sectors, e.g. machines, transport equipment and instruments.

In an extension expanding the time horizon of our analysis at the cost of less granular product categories, we use SITC-4 (around 1,000 product categories) bilateral trade flows from The Atlas of Complexity and study ten-year windows over 1962-2023.

Income data. For estimation of the income gradient of goods variety and quality as well as sample splits into 'poor' and 'rich' countries we use PPP income per capita from the Penn World Tables (PWT).

The variety price index. A conventional price index measures the cost of a *fixed* basket and is, by construction, blind to new varieties: a good, or a new source for a good, that a country did not import in the base period, has no base-period price, so a fixed-basket index registers nothing when it first appears, even though gaining access to it makes consumers better off. [Feenstra \(1994\)](#) addresses this issue by deriving the exact cost-of-living index for constant-elasticity-of-substitution (CES) preferences when the set of available varieties itself changes between periods.² For a given importer and good g with elasticity of substitution

²This is a cost-of-living index, not a gains-from-trade measure: it values the change in the set of available varieties over time from observed import shares, and isolates the variety channel. The sufficient-statistic formula of [Arkolakis et al. \(2012\)](#) instead gives the total gain from trade relative to autarky as a power of the domestic expenditure share, $\lambda_{jj}^{-1/\varepsilon}$ with trade elasticity ε , collapsing every channel into a single elasticity in general equilibrium (extended across sectors by [Costinot and Rodríguez-Clare 2014](#), with larger gains under cross-sector complementarity in [Ossa 2015](#)). Both are share-to-an-elasticity objects; the difference is that

$\sigma_g > 1$, the exact price index between the base period 0 and the end period 1 factors into two pieces,

$$\frac{P_{g,1}}{P_{g,0}} = \underbrace{\prod_{e \in C_g} \left(\frac{p_{ge,1}}{p_{ge,0}} \right)^{\omega_{ge}}}_{\text{conventional index, common varieties}} \times \underbrace{\left(\frac{\lambda_{g,1}}{\lambda_{g,0}} \right)^{1/(\sigma_g-1)}}_{\text{variety adjustment}}, \quad (3)$$

where C_g is the set of varieties (here, exporters) supplying good g in *both* periods, the ω_{ge} are ideal log-change (Sato–Vartia) weights that sum to one over C_g , and $\lambda_{g,t}$ is the share of period- t expenditure on good g that falls on the common set C_g . The first factor is an ordinary price index computed only over the varieties one can actually compare across periods; the second is the correction for the varieties that entered or exited.

All of the economics is in the second factor, and its appeal is that it needs no information on the new varieties themselves. One would seemingly need the price at which a newly available variety *would* have been priced out of the market, its reservation (virtual) price, which a complete welfare calculation appears to require. Under CES that price is formally infinite and never observed, yet it is not needed. The expenditure share retained by the surviving common varieties is a sufficient statistic for the welfare value of everything that entered or exited: CES demand awards a variety a large share only when it is attractive at its price, so when entrants capture much of end-period spending the common share $\lambda_{g,1}$ falls well below $\lambda_{g,0}$ and the index infers that the new varieties were valuable. Summing the negative log of the variety factor across goods, weighted by current-period expenditure shares w_{jg} , gives importer j 's welfare gain from the expansion of variety,

$$\ln G_j = \sum_g w_{jg} \frac{1}{\sigma_g - 1} \ln \frac{\lambda_{jg,0}}{\lambda_{jg,1}}, \quad (4)$$

in which $\lambda_{jg,t}$ now carries the importer subscript j . New entry lowers the end-period common share $\lambda_{jg,1}$ and so raises $\ln G_j$, and the gain is larger the lower is σ_g : the exponent $1/(\sigma_g - 1)$ is largest exactly when varieties are poor substitutes, so that an extra one is worth most, and it vanishes as $\sigma_g \rightarrow \infty$ and the varieties become homogeneous. To fix ideas, if the varieties that enter capture 18 percent of a good's end-period expenditure ($\lambda_{jg,1}/\lambda_{jg,0} = 0.82$) and $\sigma_g = 4$, the good's true cost of living is $0.82^{1/3} \approx 0.94$ times the level a fixed-basket index records, a 6 percent gain the conventional price index misses entirely.

Estimating the elasticities. The welfare gain depends on the elasticity of substitution σ_g , so rather than calibrate a single value we estimate it for each HS6 good by the [Feenstra \(1994\)](#) method applied to world trade, treating exporters as the competing varieties. The difficulty is the textbook one: within a good, consumer substitution across exporters traces out a downward-sloping CES demand curve while exporters' costs trace out an upward-sloping supply curve, and the data lie only at their intersection, so a single market cannot separate

we use the common-variety *import* share over time, whereas [Arkolakis et al. \(2012\)](#) use the *domestic* share against autarky. The former isolates the consumer-variety margin we measure and needs only import data.

the two. Feenstra's insight is that a panel of varieties can. Differencing each variety's price and expenditure-share changes against a reference exporter sweeps out the component common to all varieties of the good (the importer- and time-specific terms), leaving variety-specific demand and supply shocks; assuming these are mutually orthogonal, the way the dispersion of price changes co-moves with the dispersion of share changes *across* varieties identifies both elasticities, because varieties differ in the relative variance of their demand and supply shocks. That cross-variety heteroskedasticity is the identifying variation. We solve the resulting moment conditions good by good, with a grid search over the structural parameters where the closed-form solution is infeasible, following [Broda and Weinstein \(2006\)](#). The procedure yields elasticities for 4,970 HS6 goods with a median of 3.6, and 97 percent between 2 and 10. This closely matches the [Broda and Weinstein \(2006\)](#) distribution (US median near 3), which validates the implementation. The estimated σ_g enter (4) to compute importer welfare gains.

The quality tier. The intensive margin asks whether a country buys the dearer or the cheaper version of a given good. We measure it with ϕ , a trade-revealed quality index whose construction represents a within-product unit-value comparison. Let $u_{egj,t} = v_{egj,t}/q_{egj,t}$ be the unit value (value divided by quantity) of the HS6 good g that importer j buys from exporter e in year t , computed on a criteria-defined basket of everyday final-consumption goods. Separately in each year, ϕ is the importer effect in the weighted two-way fixed-effects regression

$$\ln u_{egj,t} = \delta_{eg,t} + \phi_{jt} + \varepsilon_{egj,t}, \quad (5)$$

estimated by weighted least squares with weights equal to the trade value $v_{egj,t}$, winsorised at the 99th percentile so that a few very large flows do not dominate. The work is done by the exporter-by-product fixed effect $\delta_{eg,t}$: it absorbs the price of good g as supplied by exporter e , so that ϕ_{jt} is identified purely from *within-cell* variation, i.e., from how much importer j pays for the very same good from the very same source relative to other importers. Differences in product mix across countries and in suppliers' comparative-advantage prices are thereby differenced out; what remains in ϕ_{jt} is the quality tier the importer selects within each narrow good, because an HS6 line from a single exporter still bundles cheaper and dearer variants and a higher unit value picks out the dearer, higher-specification (quality) one.

The regression is run on importers with at least 200 distinct exporter-product cells, so that the importer effect is precisely estimated, and cells containing a single observation, which carry no within-cell information, are dropped. Because a two-way fixed-effects model identifies ϕ only up to an additive constant, the index is normalised to a trusted frontier: the mean of ϕ over a set of high-income reference countries is set to zero, and each country's ϕ_{jt} is read relative to that frontier. A higher ϕ therefore means a country systematically buys the dearer, higher-quality varieties of the same goods from the same suppliers, and the change in ϕ between 1995 and 2023 is its movement up or down the consumption quality ladder. Being a unit-value object, ϕ is noisier than the variety counts; Section 5 shows the quality result is

robust across five constructions of ϕ that address this noise, including a size-adjusted version that adds a log-quantity covariate, a three-year-pooled estimate and a Kalman-smoothed series.

4 The gains from variety

4.1 The pro-poor incidence

Table 1 reports the baseline variety results. The median country saw its cost of living fall by about 15 percent through the expansion of its import variety over 1995 to 2023, several times the rich-country magnitude in Broda and Weinstein (2006) because the developing world integrated from a far narrower starting point. For the rich economies they studied the method lands in the same range: the United States gains 1.4 percent, Japan and Britain 2.7 percent respectively, compared to the 2.6 percent Broda and Weinstein report for the United States over 1972 to 2001. Developing economies gain far more (China 10 percent, India 8 percent). The income progressivity finding does not hinge on the elasticities. A common calibrated σ rescales every country's gain by the same factor $1/(\sigma - 1)$, so it moves the magnitude of the gain and its slope mechanically, but leaves the incidence unchanged: the correlation between the variety gain and initial income is -0.56 for calibrated values of 3, 5 and 8 alike, against -0.55 for the estimated σ_g (see Table 2, rows for $\sigma = 3, 5, 8$).

Table 1: Welfare gains from import variety, by income (1995–2023)

	Median gain (% cost of living)	Gradient on initial ln income (s.e.)	Correlation
All countries	15.1	-0.084 (0.012)	-0.55
Rich half	7.1	-0.036 (0.021)	-0.24
Poor half	25.4	-0.069 (0.045)	-0.23

Notes: 138 importers with population above one million. Gain is the Feenstra variety term (4) with estimated σ_g , as a cost-of-living reduction over 1995–2023. The gradient is the cross-country slope of the log gain on initial log PPP income per capita, with the Pearson correlation in the final column. Rich/poor split at the median of initial income; the gradient and correlation for each half are estimated within that half, so they use a restricted income range. These are baseline (all-flow) figures; the 'conservative' counterparts (see Table 2, dropping flows below \$50,000) used in the remainder of the paper are a gradient of -0.053 and rich/poor halves of 12 and 25 percent.

The gain falls steeply with income. The log gain regressed on initial log income has a slope of -0.084 (correlation -0.55), and the poor half of countries gained 25.4 percent against 7.1 percent for the rich half. This baseline overstates the spread, as the next paragraph demonstrates; a more conservative gradient (based on minimal reporting flows of \$50,000) is about -0.05 , with the poor half gaining roughly twice the rich half. The incidence is

progressive *within* the sample income distribution: The left panel of Figure 1 plots the variety gradient, the right panel the same for variety counts.

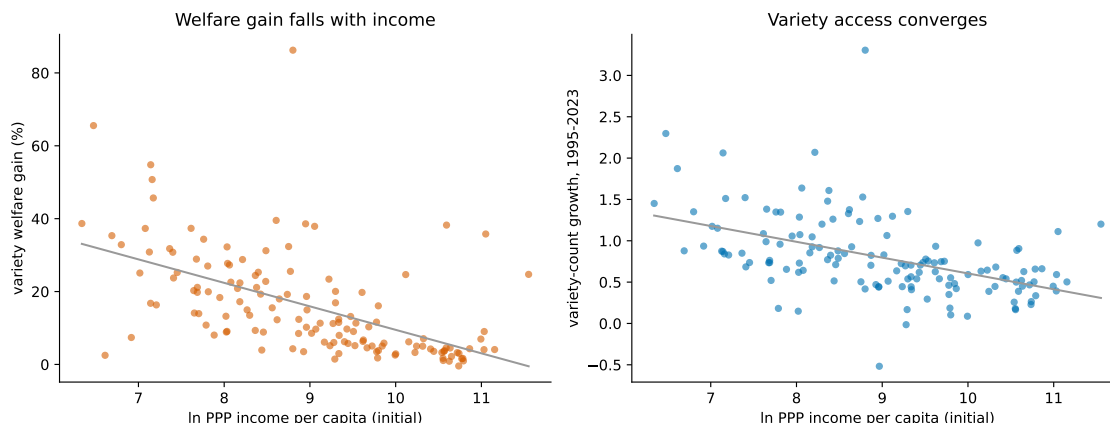


Figure 1: The variety gains were pro-poor. Left: the Feenstra welfare gain against initial income. Right: growth in the number of imported varieties (exporter-product pairs) against initial income. Both fall steeply with income. Variety count roughly doubled for the median country and grew about twice as fast in poorer than in richer economies

Robustness. Table 2 shows the gradient is not an artefact of any single modelling choice. Restricting to consumer goods, the cleanest consumer-gains object, strengthens it to -0.109 ; excluding the major entrepot exporters leaves it at -0.079 ; weighting products equally gives -0.130 ; a 2000 to 2019 window (to focus on the most recent time period but isolate the estimate from a potential Covid bias) gives a smaller but still clearly progressive -0.052 .

A natural concern, related to the debate on unconditional convergence in manufacturing (Rodrik 2013, Herrendorf et al. 2025), is that the convergence in variety could instead be carried by agricultural and other primary goods: poor countries may simply be importing more kinds of fruit! The opposite is true. Restricting to HS chapters 25 to 97 (manufactures, minerals and fuels) gives a steeper gradient of -0.095 , restricting to HS chapters 28 to 97 (chemicals onward, dropping minerals and fuels) gives -0.112 , and restricting to HS chapters 84 to 90 (machines, transport equipment and instruments) gives -0.120 . The progressive variety gradient is, if anything, stronger in manufactures than in the full sample.

Another concern is that BACI reconciles and drops very small flows, so a poor country's import that sat below the reporting floor in 1995 and rose above it by 2023 would enter as a spurious new variety, mechanically inflating poor-country gains. Dropping bilateral varieties worth less than \$10,000, \$50,000 and \$250,000 (3-year window average) gives gradients of -0.061 , -0.053 and -0.042 . The progressive incidence survives even after every sub-\$250,000 flow is removed, which retires the reporting-floor concern, though it loses about half its magnitude. The baseline gradient of -0.084 is therefore an upper bound. With sub-\$50,000 flows dropped, our preferred 'conservative' figure, the gradient is around -0.05 , with the poor half of countries gaining 25 percent against 12 percent for the rich half, a factor of

two rather than three or four. The qualitative result is robust; the precise magnitude is not, and we use the sub-\$50,000-drop figures for all computations hereafter.

Finally, these estimates apportion equal weight to each country, so that they may be distorted by an effect dominating small economies. We address this concern by including the log of initial population as an additional control in the baseline regression, resulting in a virtually identical gradient estimate.

Table 2: Robustness of the pro-poor variety gradient (1995-2023)

Variant	Median gain (%)	Gradient (s.e.)	Correlation
All goods (baseline)	15.1	−0.084 (0.012)	−0.55
Consumer goods only	14.9	−0.109 (0.013)	−0.55
Excluding entrepots	13.8	−0.079 (0.011)	−0.58
Products weighted equally	25.4	−0.130 (0.012)	−0.67
Window 2000–2019	5.4	−0.052 (0.007)	−0.41
Calibrated $\sigma = 3$	23.1	−0.129 (0.019)	−0.56
Calibrated $\sigma = 5$	11.6	−0.064 (0.009)	−0.56
Calibrated $\sigma = 8$	6.6	−0.037 (0.005)	−0.56
HS Chapters 25-97	14.7	−0.095 (0.016)	−0.53
HS Chapters 28-97	15.8	−0.112 (0.016)	−0.55
HS Chapters 84-90	10.8	−0.120 (0.024)	−0.45
Drop flows below \$10k	15.1	−0.061 (0.009)	−0.53
Drop flows below \$50k	19.4	−0.053 (0.008)	−0.49
Drop flows below \$250k	21.6	−0.042 (0.009)	−0.37
Conditional on ln population		−0.085 (0.011)	

Notes: Gain, gradient and correlation coefficient as in Table 1. The calibrated- σ rows replace the estimated elasticity of substitution σ_g with a common value; because this rescales every gain by $1/(\sigma - 1)$, the median gain and slope move mechanically with σ while the incidence is unchanged (the gain-income correlation is -0.56 throughout, against -0.55 at the estimated σ_g). Sub-sample restrictions are for manufactures, minerals and fuels (HS 25-97), manufactures and minerals (HS 28-97), and machines, transport equipment, and instruments (HS84-90). The final row additionally controls for country size (log population).

Mechanism. What drives the gradient is the starting point. Variety *levels* converged strongly: the growth in a country's *number* of imported varieties, regressed on its initial log variety count, has a slope of -0.386 (correlation -0.74), so economies that began variety-poor added proportionally far more over the sample period. Initial variety was in turn tightly linked to income, the two correlated 0.77 , because protected, autarkic, or simply poor economies imported from a narrower range. The convergence in levels was therefore progressive in incidence: the variety-deprived caught up, and they were disproportionately poor.³ This is

³The income gradient is fully mediated by this channel: adding the initial variety count to the gradient regression removes it, leaving a small positive residual ($+0.047$). That conditional coefficient is not informative

the extensive-margin analogue of income convergence, and the access margin of Section 2: as trade costs fell, poorer, variety-deprived economies gained access to the world's range of goods far faster than rich ones.

4.2 The magnitude of the gains and between-country inequality

Section 4 measured *who* gained. This section turns to how much the gains were worth in aggregate, and what they did to income inequality between countries. The variety term is a fall in the cost of living, equivalent to a gain in real consumption. The figures here use the conservative, threshold-robust variety set that drops flows below \$50,000. They are partial-equilibrium cost-of-living statistics rather than general-equilibrium welfare, and they compare across countries, so they are silent on the within-country distribution studied in [Fajgelbaum and Khandelwal \(2016\)](#). A partial check on within-country gains suggests that we do not merely capture gains for the economic elite in poor countries. Splitting consumer goods at the median world unit value, the progressive cross-country gradient is, if anything, steeper for low-tier, necessity-type goods (correlation -0.59) than for high-tier, luxury-type goods (-0.23): the variety gains reached the cheap goods poor households buy and not only the dearer ones the local elite buy.

Returning to the full variety set, the aggregate gains are sizeable. Weighting countries by population, the expansion of import variety raised real consumer welfare by an average of 14 percent over 1995 to 2023, and the gains were progressive, the poor half of countries gaining 25 percent against 12 percent for the rich half. Across the world population the progressivity compresses, with the bottom half by income gaining only somewhat more than the top, because countries with the largest populations are middle-income while the steepest gains accrue to small, very poor economies.

Being progressive, the variety gains compressed the between-country income distribution. Aggregating them lowers the population-weighted between-country Gini of income from 0.585 to 0.566, a fall of 0.019, or 3 percent, from this single channel (Figure 2). The effect is modest in absolute terms, a few percent of the Gini, but far from negligible if brought into context: the population-weighted between-country income Gini fell by about 0.13 over 1995 to 2023 as incomes converged, mostly through income growth in China and India, so the variety channel, at about 0.02, is equivalent to one sixth of that convergence, and it comes from a source the income statistics do not cover.

These numbers understate how broad-based the channel is. Population weighting lets China and India dominate that income convergence: remove these two economies and between-country income convergence over 1995 to 2023 all but disappears, the Gini moving only from 0.509 to 0.504,⁴ while the variety channel is undimmed at about 0.03 points, if anything

about a structural income effect, because initial variety is itself a mechanical determinant of the Feenstra gain, a low base leaving more room for the common-variety share to fall, and is 0.77-collinear with income, so conditioning on it partials out the convergence channel itself. It is mediation, not divergence: variety levels unambiguously converged.

⁴This is dispersion, the population-weighted between-country Gini (σ -convergence). The broad

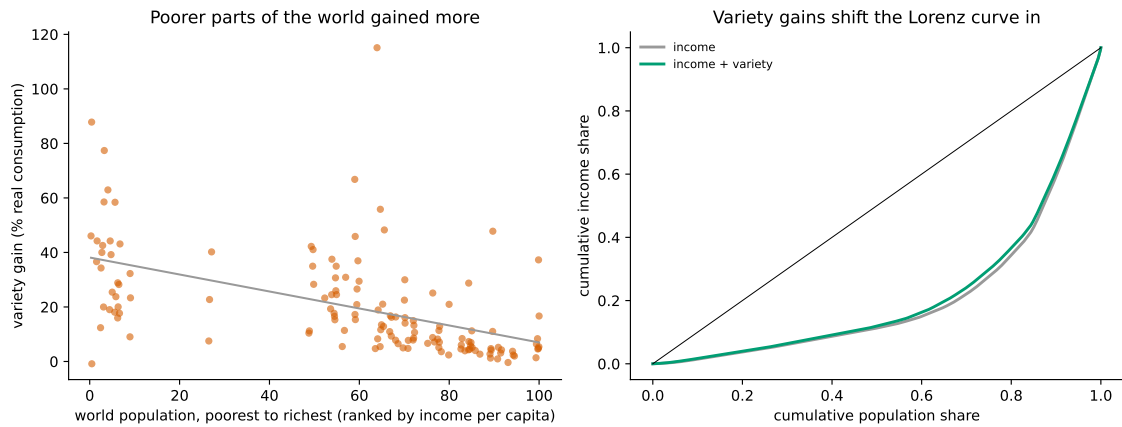


Figure 2: Variety gains narrowed inequality between countries (1995–2023). Left: a global incidence curve. The horizontal axis orders the world’s population from the poorest country to the richest by income per capita (so it is a population axis, sorted by income, not a second income axis); the vertical axis is the variety gain. Poorer parts of the world gained more. Right: the between-country Lorenz curve of income with and without the variety gains, which shift it toward the diagonal.

larger because the gains concentrate in small poor economies. For the other 136 countries, real-consumption convergence through variety was the dominant force, dwarfing a near-zero income convergence: countries grew richer in what they consumed without growing richer in income. For the median poor country the variety gain is worth several years of actual income growth, a gain in real consumption it secured through trade access alone. Whatever globalisation did to inequality through the production side, on the consumer-variety margin it was, between countries, a measurable force for convergence.

4.3 Variety convergence in the long run

The results so far cover 1995 to 2023. Is variety-access convergence specific to this era of hyperglobalisation, or a longer process? The Atlas of Economic Complexity SITC-4 bilateral data, a harmonisation of UN Comtrade flows, carry the variety *counts* back to 1962, so the convergence gradient, the slope of variety-count growth on initial income, can be estimated era by era. These historical data are value-only, so this is the extensive (count) margin rather than the full welfare index.

Over the full sixty-year span the gradient is strongly pro-poor (-0.262 , correlation -0.43), but Table 3 shows that this convergence is not a permanent, decade-by-decade feature of the data. Three patterns stand out. The 1962–1971 decade is weakly pro-poor; the 1972–1981 decade reverses sign, with rich countries adding variety faster than poor ones in both the full and the fixed 91-country samples, a pattern consistent with the oil shocks of that decade (though we do not test the channel here); and the 1980s are essentially flat. The strong, robust pro-poor gradient is concentrated in 1992–2001 and 2002–2011, the decades

unconditional convergence documented by [Patel et al. \(2021\)](#) and [Kremer et al. \(2022\)](#) is β -convergence in growth rates, a distinct measure (and see [Johnson and Papageorgiou 2020](#) for a cautious survey); our point here is only that the dispersion of income did not narrow once these two economic giants are removed.

of falling tariffs, WTO accessions and China’s emergence as a global economic power, where the fixed 91-country correlation reaches -0.41 and -0.59 respectively. The most recent period, 2012–2023, has barely any cross-country pattern at all, consistent with a slowdown in the cross-border decline of trade costs and a more contested trade-policy environment. The post-1995 result we uncover in the main analysis of the paper is therefore concentrated in two specific decades rather than spread evenly across a sixty-year trend.

Table 3: Variety (count) convergence by era (1962–2023)

Period	Median growth	Gradient (s.e.)	Corr.	N	Fixed- N (corr)
1962–2023	379%	-0.262 (0.068)	-0.43	91	-0.43
1995–2023	64%	-0.115 (0.020)	-0.38	139	-0.39
1962–1971	39%	-0.087 (0.045)	-0.25	91	-0.25
1972–1981	10%	$+0.052$ (0.034)	$+0.15$	109	$+0.18$
1982–1991	35%	-0.022 (0.018)	-0.10	114	-0.09
1992–2001	31%	-0.071 (0.017)	-0.22	139	-0.41
2002–2011	22%	-0.055 (0.006)	-0.54	144	-0.59
2012–2023	7%	-0.013 (0.009)	-0.11	148	-0.10

Notes: Variety counts are from the Atlas of Economic Complexity SITC-4 bilateral data. Endpoints are three-year averages at the start and end of each time window (‘era’). Median growth is the median country’s proportional growth in its variety count over the time window. Gradient is the slope estimate of the growth in the number of imported (SITC-4 product) varieties regressed on initial log PPP income per capita. A negative gradient indicates pro-poor variety convergence. The correlation instead just computes the Pearson correlation between variety country growth and initial log PPP income per capita. The final column is the same Pearson correlation but with the sample restricted to the 91 countries reporting in 1962–1971, an apples-to-apples sample across all time windows.

5 No convergence in quality

Convergence in the breadth of consumption need not bring convergence in its quality. A country can reach many new varieties while still buying the cheaper version of each. The intensive margin, the quality tier, is measured by ϕ , the trade-revealed index introduced in Section 3: the importer-year fixed effect from a within-product unit-value regression, which records whether a country buys the dearer or the cheaper version of the same good from the same supplier. Two features make ϕ usable here despite being noisier than the variety counts. First, its measurement error enters the convergence regression through the dependent variable, the change in ϕ regressed on initial income, where it inflates the standard error but leaves the slope unbiased, so a weak gradient is genuinely weak rather than noise hiding a strong one. Second, the finding does not depend on how ϕ is built. Across all five constructions of ϕ (the baseline, which screens out anomalous unit-value cells, representing a data-quality filter distinct from the sub-\$50,000 ‘conservative’ variety set of Section 4; a size-adjusted version

that partials out a log-quantity covariate; a size-adjusted variant in combination with the data-quality filter; a three-year-pooled estimate; and a Kalman-smoothed series), the correlations stays in a narrow band of +0.09 to +0.13, an order of magnitude weaker than the variety gradient (−0.55) and statistically indistinguishable from zero in every case. The contrast with variety is not an artefact of a particular quality index we adopt.

Table 4 reports both margins together. The variety margins are strongly progressive (correlations with initial income per capita −0.59 and −0.51).⁵ The quality margin is essentially flat in the full sample: the gradient of the *change* in ϕ on initial income is +0.020 (insignificant), the correlation 0.11: barely distinguishable from zero convergence (see the left plot in Figure 3). Yet quality did not fail to converge entirely: in a subsample of large economies, where ϕ is cleanest, it shows modest convergence, a standardised slope near −0.2. But this is less than half the variety slope for the same subsample (−0.45). The ordering holds across constructions and subsamples: poorer countries gained dramatically in variety and, at most, modestly in quality. The right plot in Figure 3 makes the contrast visible, with poor countries (orange circles) clustered at high variety gains (further to the right) and without a clear pattern for quality improvement (closer to the bottom). Of the 68 poor countries, 54 are above-median in variety gains but only 30 in quality change (the grey lines mark the respective medians).

Table 4: Two margins of consumer globalisation: income gradients (1995-2023)

Margin	Median	Gradient on income (s.e.)	Correlation
Variety welfare gain (extensive)	15.1%	−0.084 (0.012)	−0.55
Variety count growth (extensive)	107%	−0.191 (0.028)	−0.48
Quality tier $\Delta\phi$ (intensive)	−0.037	+0.020 (0.016)	+0.11

Notes: Each margin is reported on its full sample in the HS6 data: the two variety margins over 138 countries (the welfare-gain row is identical to Tables 1 and 2), the quality tier over the 136 countries with an estimate for ϕ . Restricting the variety margins to the same 136 countries as the quality margin leaves them virtually unchanged. The median is the median country's value on each margin: the welfare gain as a percentage of the cost of living, variety-count growth as a proportional change, and the quality tier as the change in the ϕ index (expressed in index points, not as a percentage). Gradient reports the cross-country slope from a regression of variety welfare gain/country growth or quality tier change on initial log PPP income per capita. The intensive (quality) margin is robust across five constructions of ϕ , with standardised slopes for the income gradient from +0.09 to +0.13 (all insignificant).

Quality is an income margin. Why did access converge while quality did not? Not because quality is unrelated to income, but because it is bound too tightly to income to converge any faster than income does. The flat convergence gradient and a strong dependence of quality

⁵The variety count growth gradient of −0.176 as well as the associate correlation of −0.51 thus indicate stronger variety convergence in this dataset of around 5,000 products than in the SITC-4 data with merely 1,000 products (−0.115 and −0.39, respectively).

on income are the same fact seen twice, once in changes and once in levels. Two facts show how tightly the quality tier is pinned to income (illustrated in Figure 4): first, quality tracks the income *level*. Across countries ϕ rises with income at a correlation of 0.78, and this is not merely a fixed national trait: in a panel regression of ϕ on log income with country and year fixed effects, the within-country slope is 0.166 ($t = 3.6$, standard errors clustered by country), almost identical to the between-country slope of 0.168. A country climbs the quality tier as it grows richer. Because ϕ holds the supplier and the product fixed, this is a demand-side choice, the dearer version of the same good from the same source; whether it reflects literal quality or instead a higher willingness to pay as income rises, income is what moves it.

Second, the *change* in quality tracks income *growth*: regressing $\Delta\phi$ on income growth and initial income gives a partial correlation of 0.34 on growth and 0.22 on the initial level. Both patterns are stable across all five constructions of ϕ , the cross-sectional correlation ranging from 0.78 to 0.82 and the growth partial from 0.34 to 0.38.

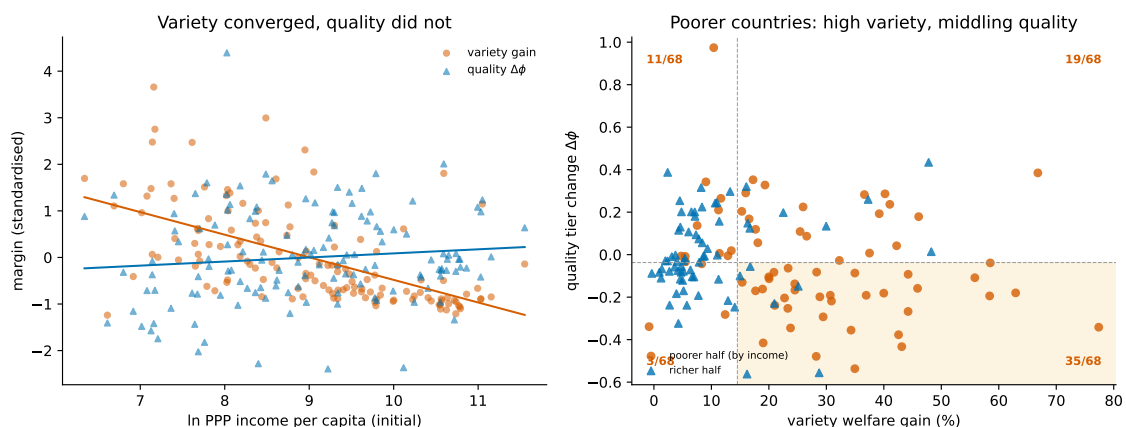


Figure 3: More choice, not more quality (1995–2023). Left: on a common (standardised) scale, the variety gain falls steeply with initial income (pro-poor) while the quality-tier change is flat. Right: the joint distribution of the two margins, with the poorer half of countries (by income) shown as orange circles and the richer half as blue triangles. The dashed grey lines mark the median of each margin; the shaded quadrant is high-variety, low-quality. The fractions report how many of the 68 poorer-half countries fall in each quadrant: they cluster to the right (54 of 68 above-median in variety) but split around the middle vertically (only 30 of 68 above-median in quality).

Together these patterns are consistent with a simple reading: quality moves with income, and income barely converged over 1995 to 2023, so neither did quality. We read this as an association, not a causal estimate, and the component of income that matters is durable income. The relationship resists clean instrumentation: the exogenous shocks that could instrument income, such as commodity-price or external-demand windfalls, move only its transitory component, whereas the quality tier tracks durable living standards, and durable cross-country income differences are not themselves exogenously assignable.⁶ We therefore characterise the relationship by triangulation, from its appearance in cross-country levels, in

⁶We nonetheless probed the link directly and could not pin it down. Instrumenting real income with the IMF commodity terms-of-trade index (Gruss and Kebhaj 2019), a fixed-weight commodity-price windfall, gives only a moderately strong first stage ($F \approx 10$), and the windfall has no detectable effect on the quality

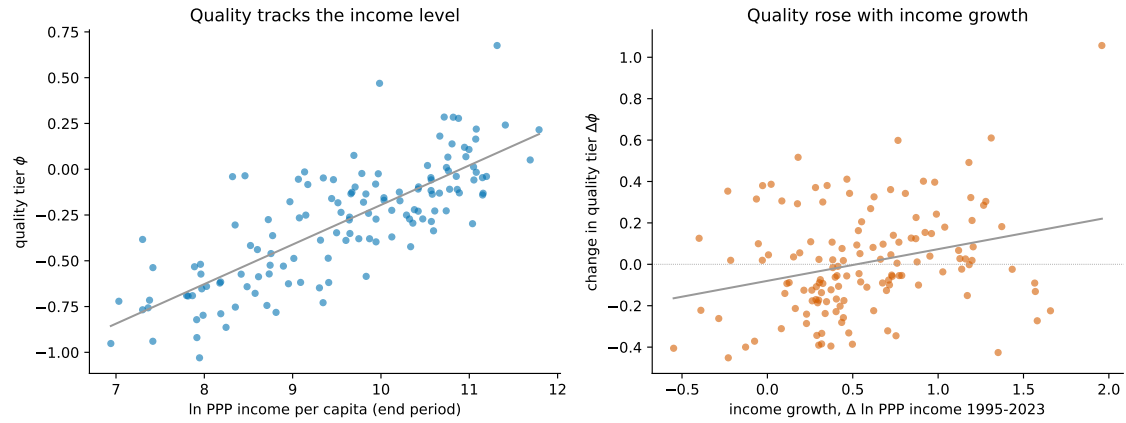


Figure 4: Quality is an income margin (1995–2023). Left: the quality tier ϕ against income per capita (level) across countries; quality rises with income at a correlation of 0.78. Right: the change in ϕ over 1995 to 2023 against income growth; countries that grew richer climbed the quality ladder. The partial correlation of $\Delta\phi$ on income growth, holding initial income fixed, is 0.34.

within-country changes, and in long-run growth, and from the fact that the income-quality relationship slopes upward exactly as non-homothetic demand predicts, with richer consumers choosing higher-quality versions of the same goods, the quality analogue of an Engel curve (Fieler 2011, Hallak 2006).

ϕ does not capture supplier identity. Another worry might be that the apparent income channel masks a supplier-choice channel: imports from advanced, technologically-rich economies might carry embodied technology, in the spirit of Grossman and Helpman (1991) and Coe and Helpman (1995), and lift the importer’s quality tier through *who* they buy from, rather than *how much* they earn. By the construction of ϕ , this alternative explanation cannot apply here: the exporter-by-product fixed effects $\delta_{eg,t}$ in equation (5) absorb every difference in offered quality across suppliers of the same good, so the Coe–Helpman channel that operates through supplier choice is, by design, in δ , not in ϕ . This holds at every level of aggregation, including in the sectors most natural for embodied-technology imports (HS 28–97 for chemicals and manufactures, HS 84–90 for machines, transport equipment and instruments). What ϕ records is the tier the importer selects *within* each supplier-product cell, and the income panel of the previous paragraph is the relevant empirical object for it. The variety gain, by contrast, is unrelated to income growth (correlation -0.006 against $+0.34$ for the quality change), so it accrued whether or not a country grew richer.

This is not to say trade cannot raise quality: it can, but the channel runs through income rather than directly through trade costs. Falling trade costs do not lift the quality tier the way they lift variety.

tier (reduced form 0.01, standard error 0.07). We do not read this as a causal zero, because the design is underpowered: the confidence interval spans both no effect and the cross-sectional association. The point estimate is small and consistent with quality responding to durable rather than transitory income, but the instrument cannot establish it. Full results are in the replication materials.

6 Conclusion

Trade reached poor consumers on one margin and not the other: variety access converged broadly and progressively, a real-consumption gain worth a median 15 percent of the cost of living that the income statistics feeding the income convergence debate, blind to new varieties, do not record. The quality tier did not converge: it follows income, and outside China and India income barely converged. Globalisation gave the world's poor consumers much more choice than quality.

The result adds a progressive consumer-side counterweight to the production-side account of trade's distributional record. On the variety channel alone it lowered between-country inequality by about 3 percent, roughly equivalent to a sixth of the income convergence over the period and virtually all of the income convergence that took place in the developing world outside China and India.

There are two natural extensions to our analysis. The first is a within-country treatment of the kind [Fajgelbaum and Khandelwal \(2016\)](#) pioneered: the necessity-versus-luxury split we apply above is reassuring but only partial, and a general-equilibrium account of how the variety gains were shared *within* poor countries is an obvious next step. The second is the *supplier-side* margin in quality, which our index sets aside by design. Because ϕ holds the supplier fixed, it captures the quality tier a country chooses *from a given source*, not the quality embodied in *which* sources it buys from. Whether importing from technologically advanced economies raises quality through embodied knowledge, in the spirit of [Coe and Helpman \(1995\)](#), is a separate question that ϕ is not designed to answer. Addressing it directly would require a different, companion index built on the variation in unit values *across* suppliers, rather than the within-supplier variation that ϕ isolates.

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